Title

AI Circuit Breaker

Authors

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Abstract

Trust in engineered systems is usually accompanied by calibration certificates and stated uncertainty budgets. For example, ECG monitors can be trusted to assess human heart health because their signals trace to measurement standards and their error bounds are known. AI systems offered to do similar safety-critical work typically come with neither. Systems engineering (SE) already supplies this kind of measurement rigor to biomedical devices, aerospace, and manufacturing. The discipline is called metrology: the engineering of measurement with stated uncertainty. AI is the exception: calibration practice is minimal, uncertainty budgets are rarely stated, and traceability to reference standards is limited.

The existing AI-trust literature contributes three strands: policy and standards frameworks, assurance-case templates and runtime monitors, and uncertainty-quantification surveys. None, to our understanding, combines these into a single continuous calibrated measurement with a traceable reference and stated uncertainty inside the SE venue. SE is being asked to field AI faster, not slower. Without a measurement basis, the speed-versus-safety tradeoff is unpriced: teams either move slow waiting for confidence they cannot quantify, or move fast and absorb unquantified risk.

We treat AI trustworthiness as a measurement problem built from three research traditions: systems-theoretic morphisms (structure-preserving mappings between a model and what it models); the Guide to the Expression of Uncertainty in Measurement (GUM) [JCGM 100:2008]; and statistical process control (SPC) [Montgomery]. The AI Circuit Breaker extends two lines of our prior work: the pairing of Bayesian methods with systems theory for test and evaluation of learning-based systems [Wach et al., INSIGHT 2022], and the isomorphic-degradation framework for morphism quality [Wach et al., CSER 2026]. These precursors are composed here into a single instrument with three stated objectives.

Objectives

The AI Circuit Breaker has three program objectives.

Objective 1: Establish morphism quality as the trust measurand. The primary measurand is morphism quality: how well the AI's internal picture of the world matches the world itself. Measurement supplies the evidence base for human judgment, not the decision.

Objective 2: Build an enforcement architecture. Turn the measurement into action, not just a dashboard readout.

Objective 3: Add a bio-inspired regulation layer. Keep the instrument from degrading silently over time.

In combination, these three objectives compress the "when is it safe to field" decision cycle, because trust becomes priceable in units decision-makers can use. The next section operationalizes them through a five-layer architecture.

Methodology: A Five-Layer Measurement Architecture

The AI Circuit Breaker is a five-layer measurement architecture (Figure 1). Each layer performs a specific metrological function, and the feedback arrows between layers encode the closed-loop logic that keeps the instrument calibrated and the operator informed. Together they operationalize the three objectives.

Layer 1, Reference Standards, holds calibrated specifications of valid domain behavior. Domain ontologies serve as gauge blocks: version-controlled, uncertainty-bounded, and recalibrated on schedule.

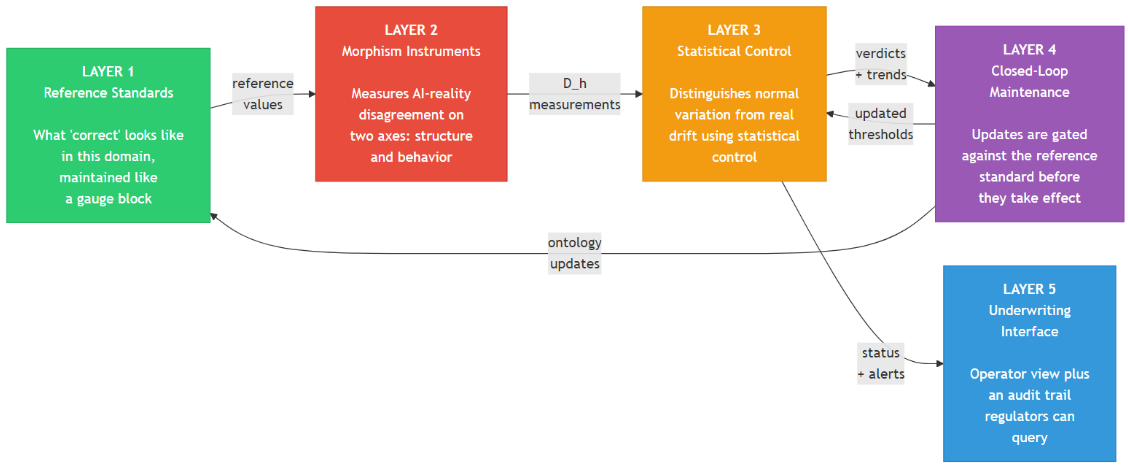

Figure 1: Five-layer architecture of the AI Circuit Breaker. Layer 1 supplies reference standards of how a domain should behave. Layer 2 measures AI-to-reality disagreement on two axes (structure and behavior). Layer 3 distinguishes normal variation from concerning drift using statistical control. Layer 4 closes the loop by gating updates against the reference standard. Layer 5 supports operator underwriting and a regulator-queryable audit trail.

Layer 2, Morphism Instruments, measures AI-reality disagreement on two axes, structural and behavioral, yielding the two-axis morphism distance. MTBH (mean time between hallucinations) summarizes longitudinal reliability. Each instrument carries a GUM-style uncertainty budget separating statistical (Type A) from prior-knowledge (Type B) sources.

Layer 3, Statistical Control, turns measurements into verdicts. Thresholds are data-derived through SPC rather than guessed; Western Electric rules detect drift; a graduated response ranges from normal autonomy through caution, restriction, and halt to full human takeover.

Layer 4, Closed-Loop Maintenance, keeps the instrument itself calibrated. Learning pipelines feed updates back into the AI, but every update is gated by an ontology-enhancement check, SHACL (Shapes Constraint Language) and SPARQL (the W3C graph query language) validation against Layer 1, so flagged failures cannot re-enter training. Bio-inspired homeostatic control maintains the operating envelope against steady-state drift; allostatic control tightens limits ahead of known high-risk windows.

Layer 5, Underwriting Interface, presents the evidence to humans. Operator dashboards show SPC chart state and MTBH reliability; a PROV-O (W3C Provenance Ontology) provenance model produces an audit trail queryable via SPARQL, giving regulators access to the provenance of any single decision.

Information reaches the decision through a chain of morphisms (physical system through sensor, digital signal, processing, features, to final decision), with chain-level bounds for morphism distance developed in [Wach et al., CSER 2026]. The formal semantic backbone is specified in the work-in-progress Circuit Breaker Trust Ontology (CBTO), an OWL 2 DL (Web Ontology Language 2, Description Logic profile) design with 25 classes, 20 object properties, 12 data properties, 6 SHACL validation shapes, and 10 SPARQL competency queries, plus PROV-O provenance.

Phased Application Spectrum

We plan three phases on safety-critical classification testbeds, each testing a specific claim of the framework.

Phase I: Single-agent morphism monitoring. Prove that morphism quality can be measured continuously on one working AI classifier in a safety-critical domain. Scope includes one learned classifier, one domain ontology, and a baseline comparison against a conventional classifier operating without the Circuit Breaker layer.

Phase II: End-to-end composition and bio-inspired regulation. Extend the monitoring to the full observation chain (classification through risk alert to safe-state), demonstrate that the chain-level composition bounds hold in practice, and add the regulation layer that keeps the Circuit Breaker itself from degrading. Homeostatic and allostatic control operate on the instrument; flagged cases are excluded from future training data.

Phase III: Domain-transfer cost analysis. Rather than build a second full implementation, quantify what it would cost to port the framework to a second safety-critical domain. Scope includes instance-data effort, domain ontology construction, sensor-to-instrument mapping, and SPC re-baselining timeline. The output is a cost estimate enterprises can apply before committing to adoption in a new domain.

Testbed selection and access status are specified in a companion research-program document. Expected performance targets across the three phases follow.

Expected Outcomes

The three phases target the following outcomes.

Phase I targets are a hallucination true-positive rate above 95 percent at 5 percent false-positive rate, classification performance at or above the domain state-of-the-art (the Circuit Breaker must not degrade base accuracy), cold-start graduation within 25 SPC subgroups (the amount of baseline data needed before the Circuit Breaker can issue reliable verdicts), and end-to-end decision latency under 25 milliseconds.

Phase II tightens these and adds end-to-end claims: hallucination detection at 97 percent, SPC-based adversarial robustness above 95 percent, and empirical confirmation of the chain-level composition bounds across the full pipeline.

Phase III delivers a quantified cost estimate for porting the framework to a second safety-critical domain, giving enterprises a concrete basis for adoption decisions.

The traceability chain, the GUM-style uncertainty budgets, and the work-in-progress CBTO specification are standalone outputs in their own right, independent of any single domain demonstrator.

Relevance to Practice

SPC and metrology are already the vocabulary of regulated industries: biomedical, aerospace, and manufacturing. The Circuit Breaker translates AI safety into their existing mental models rather than asking them to learn new AI-safety jargon. Speed comes from letting people use tools they already know.

Three practical consequences follow. Engineers gain a decision-support instrument that expresses trust as a confidence interval rather than a binary pass or fail. Regulators gain a PROV-O audit trail that makes every AI decision's provenance queryable, directly relevant to regulatory review processes across regulated domains. Program managers gain a cold-start protocol that ends deployment-readiness guesswork by quantifying how much operational data the system needs before it can be trusted.

The framework also aligns with the Visibility and Analytics pillar of Zero Trust architectures, elaborated in recent NSA guidance [NSA, 2024]. The Circuit Breaker operationalizes continuous verification of system behavior for AI components, accelerating the path to Zero Trust compliance for defense and regulated-industry deployments.

The contribution situates with respect to the INCOSE AI4SE research roadmap [McDermott et al., INSIGHT 2020] and extends its test-and-evaluation and runtime-assurance directions. The key advancements underlying these contributions are summarized below.

Key Advancements

The primary contribution is the composition of three research traditions, systems-theoretic morphisms, GUM-based measurement uncertainty, and statistical process control, into a single measurement instrument for AI trustworthiness, situated inside the SE venue where, to our understanding, the SPC dimension has not previously been represented. Specific advancements include:

- Two-axis morphism distance (structural and behavioral), building on [Wach et al., CSER 2026], as a measurable and composable quality metric for AI trust
- Composition result for observation-mediated morphism chains, extending the two-axis bounds from [Wach et al., CSER 2026]
- GUM-style uncertainty propagation applied end-to-end to AI trust measurement inside the SE venue
- SPC-derived control limits replacing intuited thresholds, paired with a defined cold-start protocol
- CBTO formal semantic backbone (OWL 2 DL + SHACL + SPARQL + PROV-O)
- Closed-loop morphism maintenance with bio-inspired homeostatic and allostatic regulation

The result is speed and innovation delivered together: speed, because trust measurement turns deployment-readiness into a priceable engineering decision instead of a committee debate; innovation, because SE metrology now extends to AI components as it already extends to every other subsystem a safety-critical system buys.

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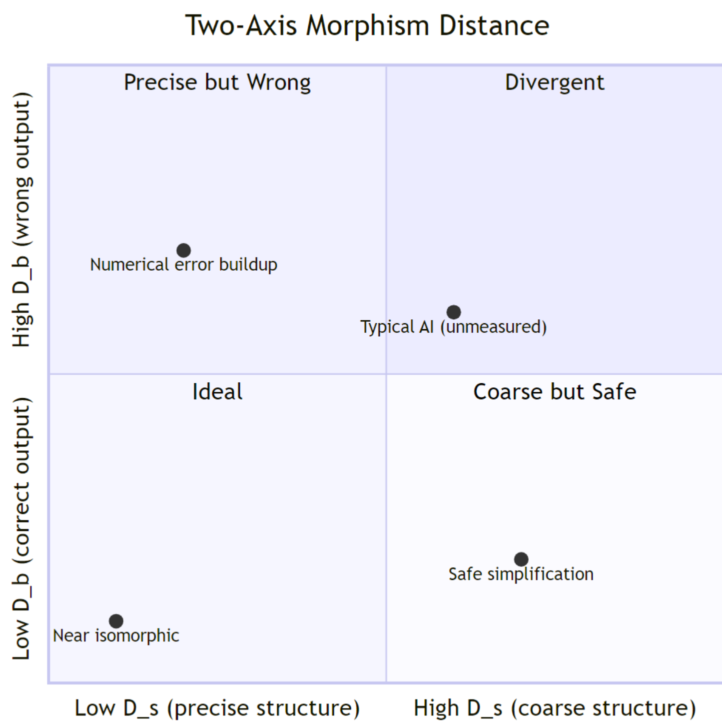

Figure 2: Two-axis morphism distance. D_s (structural distance, x-axis) captures how many distinct real-world states the AI lumps together. D_b (behavioral distance, y-axis) captures worst-case output discrepancy. Both are zero at an exact isomorphism. Four reference points illustrate typical positions: near-isomorphic (Ideal), numerical error buildup (Precise but Wrong; "discretization failure" in the formal literature), unmeasured AI (Divergent), and safe simplification (Coarse but Safe; "lumped-parameter model" in the formal literature).